

Midterm Project — Code Labs Briefs



Option A: Disaster Preparedness & Relief Info Portal

1. Project Overview & Scenario

During natural disasters or local emergencies, municipal agencies and community response teams require a lightweight, zero-dependency web utility that loads rapidly on mobile devices under low-bandwidth conditions. You are commissioned to build a client-side emergency portal that empowers citizens to locate critical contact numbers, dynamic emergency supply checklists, and evacuation zone status updates in real time.

2. Learning Objectives & Skill Alignment

- **HTML5:** Semantic architecture (<header>, <main>, <nav>, <section>, <footer>, <article>).
- **CSS3:** Responsive layout using CSS Grid and Flexbox; accessibility (WCAG AA compliance, high-contrast UI).
- **Vanilla JavaScript:**
 - Dynamic DOM filtering and searching (Array methods: filter(), map(), includes()).
 - Interactive UI manipulation (modal overlays, checklist state toggles).
 - Data persistence using browser localStorage.
- **Git & Web Hosting:** Version control workflow, deployment on GitHub Pages or Vercel.

3. Core Requirements & Features

Feature 1: Interactive Emergency Contact Directory & Filter

- **UI Structure:** Display cards for emergency services (e.g., Red Cross, Local Fire Dept, FEMA, Utility Hotlines, Animal Shelter).
- **Functionality:**
 - Provide filter buttons or a dropdown menu to filter by category (*Medical, Shelter, Utilities, Rescue*).
 - Include a quick "Copy Phone Number" button on each card that writes the number to the user's clipboard and provides visual feedback (e.g., toast notification or button text change).

Feature 2: Dynamic Supply Checklist with Local Storage

- **UI Structure:** A checklist broken into core categories (Food & Water, First Aid, Documentation, Tools).
- **Functionality:**
 - Allow users to check/uncheck recommended items.
 - Allow users to add custom checklist items via a form input.
 - Provide a progress bar showing completion percentage (e.g., "65% Prepared").
 - Save state in localStorage so checklist progress persists on page reload.

Feature 3: Evacuation Zone Search & Status Finder

- **UI Structure:** A search bar with real-time feedback and a list/grid of evacuation zones.
- **Functionality:**
 - Users can search by neighborhood name or ZIP code.
 - Filter status levels dynamically (*Mandatory Evacuation, Warning, Safe / Clear*).
 - Display color-coded badges matching the alert level (Red, Amber, Green).

4. Technical Constraints & Submission

- **No External Frameworks:** Pure Vanilla JS only (no React, jQuery, Bootstrap, or Tailwind).
- **Deployment:** Must be hosted live on GitHub Pages or Vercel.
- **Video Walkthrough:** 3 to 5-minute screencast demonstrating responsive design, DOM features, and code structure.

5. Evaluation Rubric

Criteria	Weight	Requirements
HTML/CSS Layout	30%	Fully responsive (Mobile, Tablet, Desktop) via CSS Grid/Flexbox; semantic elements used properly.
JS DOM & Functionality	35%	Working dynamic filter, live search, localStorage checklist persistence, clipboard copy API.
UI/UX & Accessibility	15%	High contrast, clean visual hierarchy, keyboard accessible, ARIA labels on controls.
Git & Deployment	10%	Clean commit history (min 8 commits), live deployed link on Vercel/GitHub Pages.
Video Walkthrough	10%	Clear walkthrough explaining technical choices and demonstrating all features.